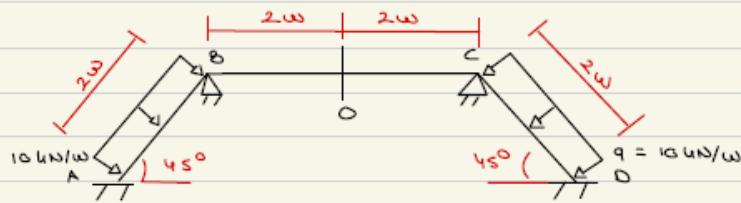
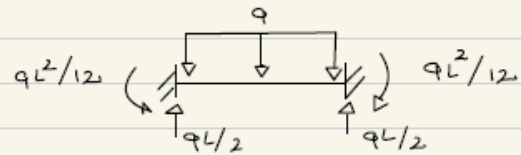


Ejercicio 3 IE. Curso 2025 - 2026. Convocatoria Ordinaria.



$$EI = 1 \cdot 10^4 \text{ kN} \cdot \text{w}^2$$

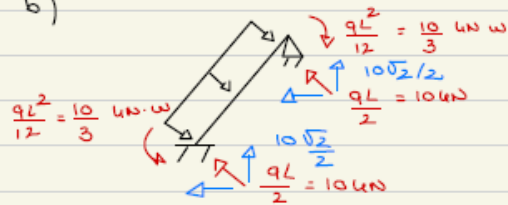
$$EA = 1 \cdot 10^5 \text{ kN}$$



a)

$$\begin{pmatrix} u_{11}^{AB} & u_{12}^{AB} & 0 \\ u_{21}^{AB} & u_{22}^{AB} + u_{11}^{BC} & u_{12}^{BC} \\ 0 & u_{21}^{BC} & u_{22}^{BC} \end{pmatrix}$$

b)



$$\{F\} = \begin{pmatrix} R_{xA} \\ R_{yA} \\ M_A \\ R_{xB} \\ R_{yB} \\ 0 \\ R_{xO} \\ 0 \\ M_O \end{pmatrix} = \begin{pmatrix} -10\sqrt{2}/2 \\ 10\sqrt{2}/2 \\ 10/3 \\ -10\sqrt{2}/2 \\ 10\sqrt{2}/2 \\ -10/3 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} R_{xA} + 10\sqrt{2}/2 \\ R_{yA} - 10\sqrt{2}/2 \\ M_A - 10/3 \\ R_{xB} + 10\sqrt{2}/2 \\ R_{yB} - 10\sqrt{2}/2 \\ 10/3 \\ R_{xO} \\ 0 \\ M_O \end{pmatrix}$$

$$c) \begin{pmatrix} RxB + 10\sqrt{2}/2 \\ RyB - 10\sqrt{2}/2 \\ uA - 10/3 \\ RxB + 10\sqrt{2}/2 \\ RyB - 10\sqrt{2}/2 \\ 10/3 \\ RxB \\ 0 \\ uA \end{pmatrix} = \begin{pmatrix} k_{11}^{AB} & k_{12}^{AB} & 0 \\ k_{21}^{AB} & k_{22}^{AB} & 0 \\ 0 & 0 & k_{11}^{BO} + k_{12}^{BO} \\ 0 & 0 & k_{21}^{BO} + k_{22}^{BO} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ \theta B \\ 0 \\ 0 \\ u_0 \\ 0 \\ 0 \end{pmatrix}$$

→ No hace falta girarla, ya que nos interesa solo el término (3,3)

$$k_{22}^{AB} + k_{11}^{BO} = \begin{pmatrix} x & x & x \\ x & x & x \\ x & x & \frac{4EI}{L} \end{pmatrix} + \begin{pmatrix} x & x & x \\ x & x & x \\ x & x & \frac{4EI}{L} \end{pmatrix} = \begin{pmatrix} x & x & x \\ x & x & x \\ x & x & \frac{8EI}{L} \end{pmatrix}$$

$$k_{12}^{BO} = \begin{pmatrix} x & x & x \\ x & x & x \\ x & -\frac{6EI}{L} & x \end{pmatrix}$$

$$k_{21}^{BO} = \begin{pmatrix} x & x & x \\ x & x & -\frac{6EI}{L} \\ x & x & x \end{pmatrix}$$

$$k_{22}^{BO} = \begin{pmatrix} x & x & x \\ x & \frac{12EI}{L^3} & x \\ x & x & x \end{pmatrix}$$

$$\begin{pmatrix} 9L^2/12 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{8EI}{L} & -\frac{6EI}{L^2} \\ -\frac{6EI}{L^2} & \frac{12EI}{L^3} \end{pmatrix} \begin{pmatrix} \theta B \\ u_0 \end{pmatrix}$$

$$\begin{pmatrix} \frac{10}{3} \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \cdot 10^4 & -1,5 \cdot 10^4 \\ -1,5 \cdot 10^4 & 1,5 \cdot 10^4 \end{pmatrix} \begin{pmatrix} \theta_B \\ V_0 \end{pmatrix}$$

d)

$$\frac{10}{3} = 1 \cdot 10^4 (4 \theta_B - 1,5 V_0)$$

$$\frac{qL^2}{12} = \frac{EI}{L} (8 \theta_B - \frac{6}{L} V_0)$$

$$0 = 1 \cdot 10^4 (-1,5 \theta_B + 1,5 V_0)$$

$$0 = \frac{EI}{L} (-\frac{6}{L} \theta_B + \frac{12}{L^2} V_0)$$

$$\theta_B = V_0$$

$$2,5 \cdot 10^4 \cdot \theta_B = \frac{10}{3} ;$$

$$\theta_B = 1,33 \cdot 10^{-4} \text{ rad}$$

$$V_0 = 1,33 \cdot 10^{-4} \text{ m}$$

$$\theta_B = V_0$$

$$V_0 = \frac{qL^4}{120EI}$$

$$\theta_C = -1,33 \cdot 10^{-4} \text{ rad}$$

$$c) \begin{pmatrix} R_{xA} \\ R_{yA} = 10 \\ M_A = 10/3 \end{pmatrix} = \begin{bmatrix} & \Delta_B \\ k_{12} & \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ \theta_B \end{pmatrix}$$

$$k_{12}^{\Delta_B} = T \cdot k_{12}^{\Delta_B'} \cdot T^T$$

$$\alpha = 45^\circ \quad T = \begin{pmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$u_{12}^{\Delta B} = T \cdot u_{12}^{\Delta B'} \cdot T^T =$$

$$= \begin{pmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} -\frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} \end{pmatrix} \cdot \begin{pmatrix} \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -\sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\sqrt{2}}{2} \left(-\frac{EA}{L}\right) & \frac{\sqrt{2}}{2} \cdot \frac{12EI}{L^3} & -\frac{\sqrt{2}}{2} \frac{6EI}{L^2} \\ \frac{\sqrt{2}}{2} \left(-\frac{EA}{L}\right) & -\frac{\sqrt{2}}{2} \cdot \frac{12EI}{L^3} & \frac{\sqrt{2}}{2} \frac{6EI}{L^2} \\ 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \end{pmatrix} \cdot \begin{pmatrix} \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -\sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} X & X & -\frac{\sqrt{2}}{2} \frac{6EI}{L^2} \\ X & X & \frac{\sqrt{2}}{2} \frac{6EI}{L^2} \\ X & X & \frac{2EI}{L} \end{pmatrix}$$

$$\begin{pmatrix} R_{xA} \\ R_{yA} - 10 \\ M_A - 10/3 \end{pmatrix} = \begin{pmatrix} X & X & -\frac{\sqrt{2}}{2} \cdot 1,5 \cdot 10^4 \\ X & X & \frac{\sqrt{2}}{2} \cdot 1,5 \cdot 10^4 \\ X & X & 1 \cdot 10^4 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1,33 \cdot 10^4 \end{pmatrix}$$

$$R_{xA} = -1,41 \text{ kN} ;$$

$$R_{yA} - 10 = 1,41 ;$$

$$M_A - 10/3 = 1,33 ;$$

$$R_{xA} = 1,41 \text{ kN}$$

$$R_{yA} = 11,41 \text{ kN}$$

$$M_A = 4,66 \text{ kN/m}$$

$$R_{xC} = -1,41 \text{ kN}$$

$$R_{yC} = 11,41 \text{ kN}$$

$$M_C = -4,66 \text{ kN/m}$$